

f-plane

Na dinâmica dos fluidos geofísicos, a aproximação do f-plane é uma aproximação onde o parâmetro de Coriolis, denotado f , é ajustado para um valor constante.

Essa aproximação é freqüentemente usada para a análise de ciclones tropicais altamente idealizados. O uso de um parâmetro constante de Coriolis evita a formação de beta-giros que são amplamente responsáveis pela direção norte-oeste da maioria dos ciclones tropicais.

Plano beta $f=f_0 + \beta y$

Na dinâmica dos fluidos geofísicos, uma aproximação pela qual o parâmetro de Coriolis, f , é ajustado para variar linearmente no espaço é chamado de aproximação do plano beta.

Em uma esfera rotativa como a Terra, varia com o seno de latitude; na assim chamada aproximação f-plane, essa variação é ignorada, e um valor de f apropriado para uma determinada latitude é usado em todo o domínio.

3- Ondas de Gravidade Inercial e distribuição de Variáveis

Nesta seção nos discutiremos o efeito da diferença centrada no espaço sobre as ondas de gravidade. Assim, nos consideramos o sistema de equação linearizada.

$$\frac{\partial u}{\partial t} = -g \frac{\partial h}{\partial x} + f v \quad 3.1a$$

$$\frac{\partial v}{\partial t} = -g \frac{\partial h}{\partial x} - f u \quad 3.1b$$

$$\frac{\partial h}{\partial t} = -H \nabla \cdot \vec{V} \quad 3.1c$$

Esta equação difere daquela da seção 2 no termo de coriolis f . O termo de coriolis não contém derivadas. Entretanto, eles são difíceis de calcular sobre a grade C, que foi ideal para ondas de gravidade puras.

Assim, nos reconsideramos o problema da distribuição de variáveis.

Não é óbvio como nos podemos analisar vários arranjos de variáveis. Nossa primeira opção é considerar (eq. 3.1) como parte de um sistema completo de equações primitivas. Nos estamos interessados no movimento de grande escala, por outro lado nos não devemos incluir o termo de Coriolis.

Sobre a grande escala, a equação primitiva admite dois tipos movimento distintos: baixa frequência e quase-geostrofico e escoamento quase-não divergente; e alta frequência ondas de gravidade inercial. Ondas de gravidade inercial são continuamente excitada na atmosfera, entretanto, como elas são dispersiva, uma acumulação local de energia de ondas dispersa com o tempo. Estes processos é conhecido como ajustamento geostrofico; o movimento permanente é um balanço aproximadamente geostrofico e muda somente lentamente com o tempo. Neste capítulo nos estaremos concentrado como simulação correta destes processos, em que é essencialmente governado pela equação de ondas de gravidade inercial(3.1).

Nos estamos interessado em ambas ondas causadas pelo efeito físico, e que é causado por inadequados dados iniciais e procedimento numérico.

Entretanto o detalhes do processo de ajustamento não importa tanto quanto a correção do resultado do escoamento quase-geotrofico.

Nos devemos no entanto investigar o efeito da distribuição do espaço de variáveis dependentes sobre a propriedade dispersiva da ondas de gravidade inercial. Este será feito usando a mais simples aproximação centrada para a derivada no espaço deixando a derivada no tempo em sua forma diferencial.

A discussão é baseada sobre aquilo que Wininghoff e Arakawa como apresentado por Arakawa (Arakawa, 1972; Arakawa et al. 1974).

Nos consideramos 5 caminhos de distribuição de variáveis dependentes. No espaço.

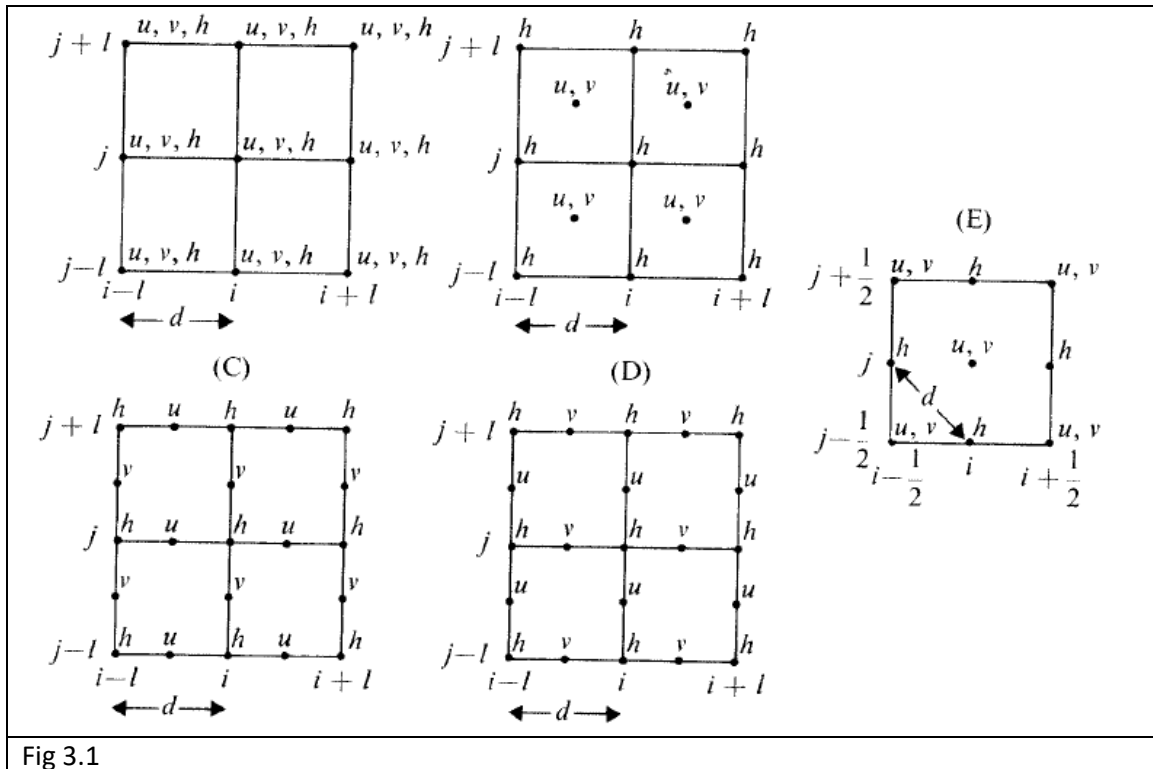


Fig 3.1

Nos Definimos d a distancia mais curta entre os pontos vizinhos carregando a mesma variável dependente. Na figura 3.1 d é o mesmo para cada uma das cinco grades. Assim, todas as

grades tem o mesmo numero de variáveis dependentes por unidade de area. A tempo de computação necessário para um integração sobre cada uma das grade será sobre a mesma; propriedade da solução obtida embora , será diferente devido ao efeito do espaço de arrajamento das variáveis.

Usando o subscripts mostrado na figura 3.1, nos definimos um operador para a diferenciação no espaço centrado.

$$(\delta_x \alpha)_{i,j} \equiv \frac{1}{d'} \left(\alpha_{i+\frac{1}{2},j} - \alpha_{i-\frac{1}{2},j} \right)$$

Esta rotação é aplicável a todas as grades. Aqui d' é a distancia entre os pontos os quais a diferença finita é feita. Assim, para a grade A, embora D d' pe igual ao tamanho da grade d , e para a grade E é igual a $\sqrt{2}d$.

Nos também definimos uma media sobre o mesmo dois pontos por:

$$(\bar{\alpha}^x)_{i,j} \equiv \frac{1}{2} \left(\alpha_{i+\frac{1}{2},j} + \alpha_{i-\frac{1}{2},j} \right)$$

Assim, $(\delta_y \alpha)_{i,j}$ e $(\bar{\alpha}^y)_{i,j}$ são definido no mesmo caminho, mas com respeito ao eixo y .

Finalmente,

$$(\bar{\alpha}^{xy})_{i,j} \equiv (\bar{\alpha}^{xy})_{i,j}$$

Para cada uma das 5 grades nos usamos uma aproximação centrada simples para a derivada no espaço e temos de coriolis (3.1). Obtemo-nos os diferentes sistemas:

GRADE A

$$\begin{aligned} \frac{\partial u}{\partial t} &= -g \bar{\delta}_x \bar{h}^x + f v = -g \left(\frac{1}{d'} \left(\bar{h}_{i+\frac{1}{2},j}^x - \bar{h}_{i-\frac{1}{2},j}^x \right) \right) + f v \\ &= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(h_{i+\frac{1}{2}+\frac{1}{2},j} + h_{i+\frac{1}{2}-\frac{1}{2},j} \right) - \frac{1}{2} \left(h_{i-\frac{1}{2}+\frac{1}{2},j} + h_{i-\frac{1}{2}-\frac{1}{2},j} \right) \right) \right) + f v \\ &= -g \left(\frac{1}{d'} \left(\frac{1}{2} (h_{i+1,j} + h_{i,j}) - \frac{1}{2} (h_{i,j} + h_{i-1,j}) \right) \right) + f v \\ &= -g \left(\frac{1}{d'} \left(\frac{1}{2} (h_{i+1,j} + h_{i,j} - h_{i,j} - h_{i-1,j}) \right) \right) + f v \\ &= -g \left(\frac{1}{2d'} (h_{i+1,j} - h_{i-1,j}) \right) + f v \end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g\overline{\delta_y}h^y - fu = -g\left(\frac{1}{d'}\left(\bar{h}_{i,j+\frac{1}{2}}^y - \bar{h}_{i,j-\frac{1}{2}}^y\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i,j+\frac{1}{2}+\frac{1}{2}} + h_{i,j+\frac{1}{2}-\frac{1}{2}}\right) - \frac{1}{2}\left(h_{i,j-\frac{1}{2}+\frac{1}{2}} + h_{i,j-\frac{1}{2}-\frac{1}{2}}\right)\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}(h_{i,j+1} + h_{i,j}) - \frac{1}{2}(h_{i,j} + h_{i,j-1})\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}(h_{i,j+1} + h_{i,j} - h_{i,j} - h_{i,j-1})\right)\right) - fu \\
&= -g\left(\frac{1}{2d'}(h_{i,j+1} - h_{i,j-1})\right) - fu
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\overline{\delta_x}u^x + \overline{\delta_y}v^y) = H\left(\left(\frac{1}{d'}\left(\bar{u}_{i+\frac{1}{2},j}^x - \bar{h}_{i-\frac{1}{2},j}^x\right)\right) + \left(\frac{1}{d'}\left(\bar{v}_{i,j+\frac{1}{2}}^y - \bar{v}_{i,j-\frac{1}{2}}^y\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(u_{i+\frac{1}{2}+\frac{1}{2},j} + u_{i+\frac{1}{2}-\frac{1}{2},j}\right) - \frac{1}{2}\left(u_{i-\frac{1}{2}+\frac{1}{2},j} + u_{i-\frac{1}{2}-\frac{1}{2},j}\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(v_{i,j+\frac{1}{2}+\frac{1}{2}} + v_{i,j+\frac{1}{2}-\frac{1}{2}}\right) - \frac{1}{2}\left(v_{i,j-\frac{1}{2}+\frac{1}{2}} + v_{i,j-\frac{1}{2}-\frac{1}{2}}\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}(u_{i+1,j} + u_{i,j}) - \frac{1}{2}(u_{i,j} + u_{i-1,j})\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}(v_{i,j+1} + v_{i,j}) - \frac{1}{2}(v_{i,j} + v_{i,j-1})\right)\right) \\
&= H\left(\frac{1}{2d'}(u_{i+1,j} - u_{i-1,j})\right) + H\left(\frac{1}{2d'}(v_{i,j+1} - v_{i,j-1})\right)
\end{aligned}$$

GRADE B

$$\begin{aligned}
\frac{\partial u}{\partial t} &= -g\overline{\delta_x}h^y + fv = -g\left(\frac{1}{d'}\left(\bar{h}_{i+\frac{1}{2},j}^y - \bar{h}_{i-\frac{1}{2},j}^y\right)\right) + fv \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}}\right) - \frac{1}{2}\left(h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) + fv \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) + fv \\
&= -g\left(\frac{1}{2d'}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right) + fv
\end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g\overline{\delta_y h^x} - fu = -g\left(\frac{1}{d'}\left(\bar{h}_{i,j+\frac{1}{2}}^x - \bar{h}_{i,j-\frac{1}{2}}^x\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} + h_{i-\frac{1}{2},j+\frac{1}{2}}\right) - \frac{1}{2}\left(h_{i+\frac{1}{2},j-\frac{1}{2}} + h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) - fu \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) - fu \\
&= -g\left(\frac{1}{2d'}\left(h_{i+\frac{1}{2},j+\frac{1}{2}} - h_{i-\frac{1}{2},j+\frac{1}{2}} + h_{i+\frac{1}{2},j-\frac{1}{2}} - h_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right) - fu
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\overline{\delta_x u^y} + \overline{\delta_y v^x}) = H\left(\left(\frac{1}{d'}\left(\bar{u}_{i+\frac{1}{2},j}^y - \bar{h}_{i-\frac{1}{2},j}^y\right)\right) + \left(\frac{1}{d'}\left(\bar{v}_{i,j+\frac{1}{2}}^x - \bar{v}_{i,j-\frac{1}{2}}^x\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}}\right) - \frac{1}{2}\left(u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}}\right) - \frac{1}{2}\left(v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= H\left(\frac{1}{2d'}\left(\left(u_{i+\frac{1}{2},j+\frac{1}{2}} - u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} - u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{2d'}\left(\left(v_{i+\frac{1}{2},j+\frac{1}{2}} - v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} - v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

GRADE C

$$\begin{aligned}
\frac{\partial u}{\partial t} &= -g\delta_x h + f\bar{v}^{xy} = -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) + f\left(\frac{1}{2}\left(\bar{v}_{i+\frac{1}{2},j}^y + \bar{v}_{i-\frac{1}{2},j}^y\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) \\
&\quad + f\left(\frac{1}{2}\left(\frac{1}{2}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}}\right) + \frac{1}{2}\left(v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) \\
&\quad + f\left(\left(\frac{1}{4}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g\delta_y h - f\bar{u}^{xy} = -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) - f\left(\frac{1}{2}\left(\bar{u}^y_{i+\frac{1}{2},j} + \bar{u}^y_{i-\frac{1}{2},j}\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) \\
&\quad - f\left(\frac{1}{2}\left(\frac{1}{2}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}}\right) + \frac{1}{2}\left(u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) \\
&\quad - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\delta_x u + \delta_y v) = H\left(\left(\frac{1}{d'}\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j}\right)\right) + \left(\frac{1}{d'}\left(v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}}\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} + v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

$$\frac{\partial u}{\partial t} = -g\left(\frac{1}{d'}\left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right)\right) + f\left(\left(\frac{1}{4}\left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)$$

$$\frac{\partial v}{\partial t} = -g\left(\frac{1}{d'}\left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right)\right) - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)$$

$$\frac{\partial h}{\partial t} = H\left(\frac{1}{d'}\left(\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} + v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}}\right)\right)\right)$$

GRADE D

$$\begin{aligned}
\frac{\partial u}{\partial t} &= -g \overline{\delta_x h^{xy}} + f \bar{v}^{xy} = -g \left(\frac{1}{d'} \left(\bar{h}_{i+\frac{1}{2},j}^{xy} - \bar{h}_{i-\frac{1}{2},j}^{xy} \right) \right) + f \left(\frac{1}{2} \left(\bar{v}_{i+\frac{1}{2},j}^y + \bar{v}_{i-\frac{1}{2},j}^y \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\bar{h}_{i+\frac{1}{2},j+\frac{1}{2}}^y + \bar{h}_{i+\frac{1}{2},j-\frac{1}{2}}^y \right) - \frac{1}{2} \left(\bar{h}_{i-\frac{1}{2},j+\frac{1}{2}}^y + \bar{h}_{i-\frac{1}{2},j-\frac{1}{2}}^y \right) \right) \right) \\
&\quad + f \left(\frac{1}{2} \left(\frac{1}{2} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} \right) + \frac{1}{2} \left(v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\bar{h}_{i+1,j}^y + \bar{h}_{i,j}^y \right) - \frac{1}{2} \left(\bar{h}_{i,j}^y + \bar{h}_{i-1,j}^y \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) = \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\bar{h}_{i+1,j}^y - \bar{h}_{i-1,j}^y \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{d'} \left(\frac{1}{2} \left(\frac{1}{2} \left(h_{i+1,j+\frac{1}{2}} + h_{i+1,j-\frac{1}{2}} \right) - \frac{1}{2} \left(h_{i-1,j+\frac{1}{2}} + h_{i-1,j-\frac{1}{2}} \right) \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right) \\
&= -g \left(\frac{1}{4d'} \left(\left(h_{i+1,j+\frac{1}{2}} - h_{i-1,j+\frac{1}{2}} + h_{i+1,j-\frac{1}{2}} - h_{i-1,j-\frac{1}{2}} \right) \right) \right) \\
&\quad + f \left(\left(\frac{1}{4} \left(v_{i+\frac{1}{2},j+\frac{1}{2}} + v_{i+\frac{1}{2},j-\frac{1}{2}} + v_{i-\frac{1}{2},j+\frac{1}{2}} + v_{i-\frac{1}{2},j-\frac{1}{2}} \right) \right) \right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial t} &= -g\overline{\delta_y}h^{xy} - f\bar{u}^{xy} = -g\left(\frac{1}{d'}\left(\bar{h}_{i,j+\frac{1}{2}}^{xy} - \bar{h}_{i,j-\frac{1}{2}}^{xy}\right)\right) - f\left(\frac{1}{2}\left(\bar{u}_{i+\frac{1}{2},j}^y + \bar{u}_{i-\frac{1}{2},j}^y\right)\right) \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{h}_{i,j+\frac{1}{2}+\frac{1}{2}}^x + \bar{h}_{i,j+\frac{1}{2}-\frac{1}{2}}^x\right) - \frac{1}{2}\left(\bar{h}_{i,j-\frac{1}{2}+\frac{1}{2}}^x + \bar{h}_{i,j-\frac{1}{2}-\frac{1}{2}}^x\right)\right)\right) \\
&\quad - f\left(\frac{1}{2}\left(\frac{1}{2}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}}\right) + \frac{1}{2}\left(u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}(\bar{h}_{i,j+1}^x + \bar{h}_{i,j}^x) - \frac{1}{2}(\bar{h}_{i,j}^x + \bar{h}_{i,j-1}^x)\right)\right) \\
&\quad - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) = \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}(\bar{h}_{i,j+1}^y - \bar{h}_{i,j-1}^y)\right)\right) \\
&\quad - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= -g\left(\frac{1}{d'}\left(\frac{1}{2}\left(\frac{1}{2}\left(h_{i+\frac{1}{2},j+1} + h_{i-\frac{1}{2},j+1}\right) - \frac{1}{2}\left(h_{i+\frac{1}{2},j-1} + h_{i-\frac{1}{2},j-1}\right)\right)\right)\right) \\
&\quad - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right) \\
&= -g\left(\frac{1}{4d'}\left(\left(h_{i+\frac{1}{2},j+1} - h_{i+\frac{1}{2},j-1} + h_{i-\frac{1}{2},j+1} - h_{i-\frac{1}{2},j-1}\right)\right)\right) \\
&\quad - f\left(\left(\frac{1}{4}\left(u_{i+\frac{1}{2},j+\frac{1}{2}} + u_{i+\frac{1}{2},j-\frac{1}{2}} + u_{i-\frac{1}{2},j+\frac{1}{2}} + u_{i-\frac{1}{2},j-\frac{1}{2}}\right)\right)\right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h}{\partial t} &= -H(\overline{\delta_x u^{xy}} + \overline{\delta_y v^{xy}}) = H\left(\left(\frac{1}{d'}\left(\bar{u}_{i+\frac{1}{2},j}^{xy} - \bar{u}_{i-\frac{1}{2},j}^{xy}\right)\right) + \left(\frac{1}{d'}\left(\bar{v}_{i,j+\frac{1}{2}}^{xy} - \bar{v}_{i,j-\frac{1}{2}}^{xy}\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{u}_{i+\frac{1}{2}+\frac{1}{2},j}^y + \bar{u}_{i+\frac{1}{2}-\frac{1}{2},j}^y\right) - \frac{1}{2}\left(\bar{u}_{i-\frac{1}{2}+\frac{1}{2},j}^y + \bar{u}_{i-\frac{1}{2}-\frac{1}{2},j}^y\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{v}_{i,j+\frac{1}{2}+\frac{1}{2}}^x + \bar{v}_{i,j+\frac{1}{2}-\frac{1}{2}}^x\right) - \frac{1}{2}\left(\bar{v}_{i,j-\frac{1}{2}+\frac{1}{2}}^x + \bar{v}_{i,j-\frac{1}{2}-\frac{1}{2}}^x\right)\right)\right) \\
&= H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{u}_{i+1,j}^y + \bar{u}_{i,j}^y\right) - \frac{1}{2}\left(\bar{u}_{i,j}^y + \bar{u}_{i-1,j}^y\right)\right)\right) \\
&\quad + H\left(\frac{1}{d'}\left(\frac{1}{2}\left(\bar{v}_{i,j+1}^x + \bar{v}_{i,j}^x\right) - \frac{1}{2}\left(\bar{v}_{i,j}^x + \bar{v}_{i,j-1}^x\right)\right)\right) \\
&= H\left(\frac{1}{2d'}\left(\bar{u}_{i+1,j}^y - \bar{u}_{i-1,j}^y\right)\right) + H\left(\frac{1}{2d'}\left(\bar{v}_{i,j+1}^x - \bar{v}_{i,j-1}^x\right)\right) \\
&= H\left(\frac{1}{2d'}\left(\frac{1}{2}\left(u_{i+1,j+\frac{1}{2}} + u_{i+1,j-\frac{1}{2}}\right) - \frac{1}{2}\left(u_{i-1,j+\frac{1}{2}} + u_{i-1,j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{2d'}\left(\frac{1}{2}\left(v_{i+\frac{1}{2},j+1} + v_{i-\frac{1}{2},j+1}\right) - \frac{1}{2}\left(v_{i+\frac{1}{2},j-1} + v_{i-\frac{1}{2},j-1}\right)\right)\right) \\
&= H\left(\frac{1}{4d'}\left(\left(u_{i+1,j+\frac{1}{2}} + u_{i+1,j-\frac{1}{2}}\right) - \left(u_{i-1,j+\frac{1}{2}} + u_{i-1,j-\frac{1}{2}}\right)\right)\right) \\
&\quad + H\left(\frac{1}{4d'}\left(\left(v_{i+\frac{1}{2},j+1} + v_{i-\frac{1}{2},j+1}\right) - \left(v_{i+\frac{1}{2},j-1} + v_{i-\frac{1}{2},j-1}\right)\right)\right)
\end{aligned}$$

$$(\delta_x \alpha)_{i,j} \equiv \frac{1}{d'} \left(\alpha_{i+\frac{1}{2},j} - \alpha_{i-\frac{1}{2},j} \right)$$

$$(\bar{\alpha}^x)_{i,j} \equiv \frac{1}{2} \left(\alpha_{i+\frac{1}{2},j} + \alpha_{i-\frac{1}{2},j} \right)$$

$$\frac{\partial u}{\partial t} = -g \delta_x h + f v = -g \left(\frac{1}{d'} \left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j} \right) \right) + f v_{i,j}$$

$$\frac{\partial v}{\partial t} = -g \delta_y h - f u = -g \left(\frac{1}{d'} \left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}} \right) \right) - f u_{i,j}$$

$$\begin{aligned} \frac{\partial h}{\partial t} &= -H(\delta_x u + \delta_y v) = H \left(\left(\frac{1}{d'} \left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} \right) \right) + \left(\frac{1}{d'} \left(v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}} \right) \right) \right) \\ &= H \left(\frac{1}{d'} \left(\left(u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j} + v_{i,j+\frac{1}{2}} - v_{i,j-\frac{1}{2}} \right) \right) \right) \end{aligned}$$

Nos podemos primeiro analisar um caso unidimensional , que na qual as variáveis u,v,h não variam com y, Assim nos temos

$$u, v, h = u(x, t), v(x, t), h(x, t)$$

A equação 3.1 se reduz a

$\frac{\partial u}{\partial t} = -g \frac{\partial h}{\partial x} + f v$ $\frac{\partial v}{\partial t} = -f u$ $\frac{\partial h}{\partial t} = -H \frac{\partial u}{\partial x}$	3.3
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Substituindo a 1,2

$$u(x, t) = Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial u}{\partial t} = -iv Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = (-iv)(-iv)Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -v^2 Re[\hat{u}e^{-i(kx-vt)}]$$

$$h(x, t) = Re[\hat{h}e^{-i(kx-vt)}]$$

$\frac{\partial^2 u}{\partial^2 t} = -g \frac{\partial}{\partial x} \frac{\partial h}{\partial t} + f \frac{\partial v}{\partial t}$ $\frac{\partial v}{\partial t} = -fu$ $\frac{\partial h}{\partial t} = -H \frac{\partial u}{\partial x}$	3.3
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$\frac{\partial^2 u}{\partial^2 t} = -g \frac{\partial}{\partial x} \left(-H \frac{\partial u}{\partial x} \right) + f(-fu)$	3.3
---	-----

$\frac{\partial^2 u}{\partial^2 t} = gH \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) - f^2 u$	3.3
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$$u(x, t) = Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial u}{\partial t} = iv Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = (iv)(iv)Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -v^2 Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial u}{\partial x} = -ik Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 x} = (-ik)(-ik)Re[\hat{u}e^{-i(kx-vt)}]$$

$$\frac{\partial^2 u}{\partial^2 x} = -k^2 Re[\hat{u}e^{-i(kx-vt)}]$$

$-v^2 Re[\hat{u}e^{-i(kx-vt)}] = -gHk^2 Re[\hat{u}e^{-i(kx-vt)}] - f^2 Re[\hat{u}e^{-i(kx-vt)}]$	3.3
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$-v^2 = -gHk^2 - f^2$	3.3
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$v^2 = gHk^2 + f^2$	3.3
$\left(\frac{v}{f}\right)^2 = \frac{gH}{f^2}k^2 + 1$	3.3

$\left(\frac{v}{f}\right)^2 = \frac{\lambda^2}{f}k^2 + 1$	3.3
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Assim, o raio de deformação :

$$\lambda = \frac{\sqrt{gH}}{f}$$

Nunca é igual a zero, a frequência da onda de gravidade inercial é uma função de k monotonicamente crescente . Portanto a velocidade de grupo $\frac{\partial v}{\partial k}$ nunca sera igual a zero.

Isto é muito importante para os processos de ajustamento geotrópico. pois impede uma acumulação local de energia das ondas.

No focaremos no efeito de diferenciação finita no espaço . As variaveis são assumidas não dependente de y , o sistema 3.2 reduz a:

$$\frac{\partial u}{\partial t} = -g\overline{\delta_x h^x} + fv$$

$$\frac{\partial v}{\partial t} = -fu$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})$$

$$\frac{\partial u}{\partial t} = -g\overline{\delta_x h^x} + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{d'}\left(\bar{h}^x_{i+\frac{1}{2},j} - \bar{h}^x_{i-\frac{1}{2},j}\right) + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{d'}\left(\frac{1}{2}\left(h_{i+\frac{1}{2}+\frac{1}{2},j} + h_{i+\frac{1}{2}-\frac{1}{2},j}\right) - \frac{1}{2}\left(h_{i-\frac{1}{2}+\frac{1}{2},j} + h_{i-\frac{1}{2}-\frac{1}{2},j}\right)\right) + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{2d'}\left((h_{i+1,j} + h_{i,j}) - (h_{i,j} + h_{i-1,j})\right) + fv$$

$$\frac{\partial u}{\partial t} = -g\frac{1}{2d'}\left((h_{i+1,j} - h_{i-1,j})\right) + fv$$

$$\frac{\partial^2 u}{\partial^2 t} = -g\frac{1}{2d'}\left(\left(\frac{\partial h_{i+1,j}}{\partial t} - \frac{\partial h_{i-1,j}}{\partial t}\right)\right) + fv$$

$$\frac{\partial^2 u}{\partial^2 t} = -g\frac{1}{2d'}\left(\left(\frac{\partial h_{i+1,j}}{\partial t} - \frac{\partial h_{i-1,j}}{\partial t}\right)\right) + fv$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{d'} \left(\bar{u}^x_{i+1+\frac{1}{2},j} - \bar{u}^x_{i+1-\frac{1}{2},j} \right) = -H \frac{1}{d'} \left(\bar{u}^x_{i+\frac{3}{2},j} - \bar{u}^x_{i+\frac{1}{2},j} \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{d'} \left(\frac{1}{2} \left(u^x_{i+\frac{3}{2}+\frac{1}{2},j} + u^x_{i+\frac{3}{2}-\frac{1}{2},j} \right) - \frac{1}{2} \left(u^x_{i+\frac{1}{2}+\frac{1}{2},j} + u^x_{i+\frac{1}{2}-\frac{1}{2},j} \right) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{2d'} \left((u^x_{i+2,j} + u^x_{i+1,j}) - (u^x_{i+1,j} + u^x_{i,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{d'} \left(\bar{u}^x_{i-1+\frac{1}{2},j} - \bar{u}^x_{i-1-\frac{1}{2},j} \right) = -H \frac{1}{d'} \left(\bar{u}^x_{i-\frac{1}{2},j} - \bar{u}^x_{i-\frac{3}{2},j} \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{d'} \left(\frac{1}{2} \left(u^x_{i-\frac{1}{2}+\frac{1}{2},j} + u^x_{i-\frac{1}{2}-\frac{1}{2},j} \right) - \frac{1}{2} \left(u^x_{i-\frac{3}{2}+\frac{1}{2},j} + u^x_{i-\frac{3}{2}-\frac{1}{2},j} \right) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{d'} \left(\frac{1}{2} (u^x_{i,j} + u^x_{i-1,j}) - \frac{1}{2} (u^x_{i-1,j} + u^x_{i-2,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i+1,j} = -H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right)$$

$$\frac{\partial h}{\partial t} = -H(\overline{\delta_x u^x})_{i-1,j} = -H \frac{1}{2d'} \left((u^x_{i,j} - u^x_{i-2,j}) \right)$$

$$\frac{\partial^2 u}{\partial^2 t} = -g \frac{1}{2d'} \left(\left(\frac{\partial h_{i+1,j}}{\partial t} - \frac{\partial h_{i-1,j}}{\partial t} \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -g \frac{1}{2d'} \left(\left(-H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right) - \left(-H \frac{1}{2d'} \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -g \frac{1}{2d'} \left(\left(-H \frac{1}{2d'} \left((u^x_{i+2,j} - u^x_{i,j}) \right) + H \frac{1}{2d'} \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -\frac{gH}{4d'^2} \left(\left(- \left((u^x_{i+2,j} - u^x_{i,j}) \right) + \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -\frac{gH}{4d'^2} \left(\left(- \left((u^x_{i+2,j} - u^x_{i,j}) \right) + \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = -\frac{gH}{4d'^2} \left(\left(- \left((u^x_{i+2,j} - u^x_{i,j}) \right) + \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = \frac{gH}{4d'^2} \left(\left(\left((u^x_{i+2,j} - u^x_{i,j}) \right) - \left((u^x_{i,j} - u^x_{i-2,j}) \right) \right) \right) + f v$$

$$\frac{\partial^2 u}{\partial^2 t} = \frac{gH}{4d'^2} \left(\left((u^x_{i+2,j} + u^x_{i-2,j}) - (2u^x_{i,j}) \right) \right) + f v$$

$(\delta_x \alpha)_{i,j} \equiv \frac{1}{d'} \left(\alpha_{i+\frac{1}{2},j} - \alpha_{i-\frac{1}{2},j} \right)$ $(\delta_x \delta_x \alpha)_{i,j} \equiv \frac{1}{d'^2} \left(\alpha_{i+1,j} + \alpha_{i-1,j} - 2\alpha_{i,j} \right)$ $(\bar{\alpha}^x)_{i,j} \equiv \frac{1}{2} \left(\alpha_{i+\frac{1}{2},j} + \alpha_{i-\frac{1}{2},j} \right)$	
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$$u(x, t) = Re[\hat{u}e^{-i(kx-vt)}]$$

$$h(x, t) = Re[\hat{h}e^{-i(kx-vt)}]$$

$$x = j\Delta x$$

$$t = m\Delta t$$

$$u(x, t) = Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$h(x, t) = Re[\hat{h}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial u}{\partial t} = -i(-v)Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -i(-mv)(-i(-mv))Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u}{\partial^2 t} = (imv)(imv)Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u}{\partial^2 t} = -(mv)^2 Re[\hat{u}e^{-i(k(j\Delta x)-v(m\Delta t))}]$$

$$\frac{\partial^2 u(j = n, m = 1)}{\partial^2 t} = -(v)^2 Re[\hat{u}e^{-i(k(n)\Delta x - v\Delta t)}]$$

$$\frac{\partial^2 u(j = n, m = 1)}{\partial^2 t} = -(v)^2 Re[\hat{u}(e^{-i(nk\Delta x - v\Delta t)})]$$

$$u(j = n, m = 1) = Re[\hat{u}e^{-i(kn\Delta x - v(\Delta t))}]$$

$$\begin{aligned} u(j = n + 2, m = 1) &= Re[\hat{u}e^{-i(k(n+2)\Delta x - v(\Delta t))}] = Re[\hat{u}e^{-i(nk\Delta x + 2k\Delta x - v(\Delta t))}] \\ &= Re[\hat{u}e^{-i(2k\Delta x)}e^{-i(nk\Delta x - v\Delta t)}] \end{aligned}$$

$$u(j = n + 2, m = 1) = Re[\hat{u}(e^{-i(2k\Delta x)}e^{-i(nk\Delta x - v\Delta t)})]$$

$$\begin{aligned} u(j = n - 2, m = 1) &= Re[\hat{u}e^{-i(k(n-2)\Delta x - v(\Delta t))}] = Re[\hat{u}e^{-i(nk\Delta x - 2k\Delta x - v(\Delta t))}] \\ &= Re[\hat{u}e^{i(2k\Delta x)}e^{-i(nk\Delta x - v\Delta t)}] \end{aligned}$$

$$u(j = n + 2, m = 1) = Re[\hat{u}(e^{i(2k\Delta x)}e^{-i(nk\Delta x - v\Delta t)})]$$

$$\begin{aligned}
& -(v)^2 \operatorname{Re}[\hat{u}(e^{-i(nk\Delta x - v\Delta t)})] \\
&= \frac{gH}{4d'^2} \left(\left(\operatorname{Re}[\hat{u}(e^{-i(2k\Delta x)} e^{-i(nk\Delta x - v\Delta t)})] + \operatorname{Re}[\hat{u}(e^{i(2k\Delta x)} e^{-i(nk\Delta x - v\Delta t)})] \right) \right) \\
& - \left(2 \operatorname{Re}[\hat{u} e^{-i(kn\Delta x - v(\Delta t))}] \right) - f^2 \operatorname{Re}[\hat{u} e^{-i(kn\Delta x - v(\Delta t))}]
\end{aligned}$$

$$-(v)^2 = \frac{gH}{4d'^2} \left(\left([(e^{-i(2k\Delta x)})] + [(e^{i(2k\Delta x)})] \right) - (2) \right) - f^2$$

$$(v)^2 = f^2 - \frac{gH}{4d'^2} \left(\left([(e^{-i(2k\Delta x)})] + [(e^{i(2k\Delta x)})] \right) - (2) \right)$$

$$\left(\frac{v}{f}\right)^2 = 1 - \frac{gH}{4f^2 d'^2} \left(\left([(e^{i(2k\Delta x)} + e^{-i(2k\Delta x)})] \right) - (2) \right)$$

$$\left(\frac{v}{f}\right)^2 = 1 - \frac{gH}{2f^2 d'^2} \left(\left(\left(\frac{e^{i(2k\Delta x)} + e^{-i(2k\Delta x)}}{2} \right) - 1 \right) \right)$$

$$\left(\frac{v}{f}\right)^2 = 1 - \frac{gH}{2f^2 d'^2} ((\cos(2k\Delta x) - 1))$$

$$\left(\frac{v}{f}\right)^2 = 1 + \frac{gH}{2f^2 d'^2} (1 - \cos(2k\Delta x))$$

$$\left(\frac{v}{f}\right)^2 = 1 + \frac{gH}{f^2 d'^2} \frac{1}{2} (1 - \cos(2k\Delta x))$$

$$\left(\frac{v}{f}\right)^2 = 1 + \frac{gH}{f^2 d'^2} \sin^2(k\Delta x)$$

$\cos x = \operatorname{Re}(e^{-ix}) = \frac{e^{ix} + e^{-ix}}{2}$	$\sin x = \operatorname{Im}(e^{-ix}) = \frac{e^{ix} - e^{-ix}}{2i}$
$\sin^2 x + \cos^2 x = 1$	$\sin^2 x = \frac{1}{2} (1 - \cos(2x))$

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